# Automated Blood Management System: Streamlining Search, Inventory, and Patient Care Information Administration

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## Abstract

*In the face of rising healthcare demands, the efficient management of blood resources has become extremely crucial. This research presents The Development of a Real-Time Online Blood Bank Management System, leveraging a stack of technologies including Python Django Framework, JavaScript, HTML, CSS, MySQL, and Agile Methodology using Adaptive Software Development (ASD) framework. The system ensures adaptability to evolving requirements, user-friendly feedback, and separation of concern for future maintenance. It also enables efficient data management and retrieval, secure storage of information, real-time inventory visibility, and streamlined request processing. It addresses the critical need for swift blood distribution and accessibility by incorporating a user-friendly web interface, allowing donors, recipients, and administrators to interact seamlessly. These technologies ensure adaptability to evolving requirements, user-friendly feedback and separation of concern for future maintenance. Incorporating a MySQL database to enable efficient data management and retrieval, ensuring secure storage of donor and recipient information, as well as blood inventory records, real-time inventory visibility to optimize blood allocation while streamlined request processing reduces response time. This approach fosters ongoing collaboration among stakeholders, promoting the system's flexibility and adaptability. The implementation of this software represents a significant step towards achieving quick and efficient blood distribution, enhancing accessibility, and ultimately saving lives.*

**Keywords:** Blood Management system, Development, Real-time, Blood bank, Management information system

## Introduction

Blood is a valuable life resource in healthcare, and the availability of safe and adequate blood supply is of high importance in saving lives during emergencies, surgeries, and other medical procedures. The Nigeria healthcare sector, despite the advancement in technology, still operates in the traditional manual method which is the paper and physical filing method. There has been inefficient in cases of emergencies due to the time consumption and human delay and poor communication when blood is urgently needed this could result in an avoidable death case, leading to more complications or even introduce another sickness or infection to the patient's health when a case of a family member has to supply their blood without proper screening because of time factor. Introducing digitalization of blood banking by designing an online blood bank management system for electronic documentation of donor and patient information with security measures in check and aiding donors and patients access to quickly locate nearby blood bank and giving blood bank administrator a real-time inventory management and centralized data storage will smooth operation and activities involve in blood banking hereby giving an efficient and effective environment for the management (Ovri, 2023). Online blood bank management systems have their origins in the late 20th century, when blood banks first began, using computerized systems for simple record-keeping and inventory control. Web-based apps started to appear in the 1990s with the development of the Internet, creating easy access to databases from various places. With the integration of donor management and recruiting functions, the development of online blood bank administration systems picked up speed in the early 2000s. These tools streamlined the blood donation procedure by enabling potential donors to register online, locate donation sites, and make appointments. Cloud-based solutions became popular in the 2010s, offering greater scalability, accessibility, and data backup options for blood banks of all sizes. The availability of blood donated records is poorly or not synchronized in the healthcare sector of Nigeria at any given point in time and the procedure of undergoing blood transfusion is not an easy one since most patients have to source blood in the event whereby the hospitals lack a needed blood type. Most patients have been forced to source blood from family as replacer donors a condition that according to the US Department of Health, (2005) is unadvisable due to a high number of infection cases. There is always the undeniable possibility of having a blood bank lack sufficient volume of some blood groups leaving patients stranded and some lives have been lost this way. The design and implementation of an online blood bank management system arises from the need to ensure the efficient and effective management of blood donation, storage, and transfusion. The term “Blood Bank” is an arm in a hospital laboratory or medical laboratory where blood products are kept and appropriate testing is carried out to reduce the health risk of transfusion-related accidents (Arif et. Al 2012). Blood bank handles and or oversees blood donations, blood tests, and transfusion and manages every blood transfusion transaction and this process is handled by the pathology department, the most important division of any hospital. It processes blood that will be supplied to the patients according to their needs. Before the blood is supplied to the patients, the blood will undergo several tests to ensure that the blood receiver is not infected by serious diseases (Hussain and Shazli 2019). The transfusion process involves data management and before now, the data management was done traditionally which was hectic and inefficient till present day only 20% of blood banks have gone digital in Nigeria and 53% globally are only private hospitals based on W.H.O statistics 2022. There is a 98% chance that in one’s lifetime, either you or someone you know will need blood.

The introduction and usage of an online blood bank management system has improved storage record keeping, centralized blood donation information and gives blood donors informative blood donation tips and patients swift access to blood supply locations in Nigeria.

**Statement of the Problem**

Blood banks play a crucial role in ensuring the availability of safe and compatible blood products for medical exercises, despite the advancement in medical healthcare technology, the majority of blood banks in Nigeria still operate using the traditional manual inventory and data management which is ineffective, inefficient, and has a prevalent problem in the availability of needed blood types. There has been a different occurring scenario where a person needs a particular type of blood and this is not available in the current hospital where a patient is being treated and leaving family members to go source for the blood type. A case scenario was a case of train incident that happened on March 9, 2022, in Lagos state where people came online to various social media platforms sourcing for donors for victims. Some key problems are;

i. **Inefficiency and Error-Prone Processes:** Traditional manual methods of record keeping and tracking of blood inventory, donor details and transfusion exercise are prone to error and are inefficient. The absence of proper synchronization of real-time data and retrieval slows down critical processes and compromises data accuracy.
ii. **Limited Accessibility and Communication:** Traditional manual method of paper-based systems hinders effective communication and coordination between blood bank staff, hospitals, and donors. This results in difficulties in matching blood types, coordinating donation events and responding to emergencies promptly.
iii. **Resource Inefficiencies and Cost:** The traditional manual method of processing blood donation exercises in a blood bank is human resource and time-consuming for valuables such as paper, ink, time etc. In addition, the cost associated with managing and maintaining a paper-based system impedes the optimal use of available resources.

**Objectives**

The study aims to develop an online blood bank management system that implements a real-time management system for efficient blood distribution and accessibility. The specific objectives of the study are to:

i. Develop a real-time inventory tracking system to monitor blood availability using MYSQL and Django Python framework.
ii. integrate authentication and access controls to ensure authorized user access using the Django authentication framework.

iii. create a platform that can be easily modifiable to incorporate new technologies and best practices.

**Significance of the Problem**

The findings and implementation of this research will benefit blood banking in ways that will enhance healthcare management;

i. by streamlining the operation involved in managing blood bank inventory, storing, distribution of blood and communication between users of the system which will result in effective blood donation.
ii. The automation processes like donor registration, patient registration, appointment scheduling and inventory management will minimize the administrative workload on blood bank staff and give room to focus on other important tasks. Thereby reducing administrative burden.

**Review of Related Literature**

The evolution of blood bank management systems from manual record-keeping to computer-based and subsequently to online solutions reflects an ongoing commitment to improving efficiency, accuracy, and accessibility in blood banking. This evolution has laid the foundation for the design and implementation of more advanced online blood bank management systems that address contemporary challenges and requirements in the healthcare landscape (Gupta and Sood 2015). The traditional blood bank management systems relied on manual record-keeping, which often resulted in inefficiencies, errors, and difficulties in maintaining an up-to-date inventory (Islam et al 2014). Despite their many benefits and significant effects, the implementation of an online blood bank management system also comes with challenges that could be discouraging to users and developers of the system (Jeyakumar 2020). Thangavel, and Senthil (2022), propose the development of a web-based blood bank management system that can store and manage information related to blood donors, recipients, blood groups, blood banks, and blood stock using PHP and MySQL as the main technologies. The system provides features such as online registration, login, search, request, and donation of blood, as well as feedback and suggestions for the users. Alshammari (2022), proposed the development of a web-based blood bank management system using the Vue.js framework to address the management system of blood banking. The system features include online registration, login, search, request, and donation of blood, as well as feedback and suggestions for users using Vuetify and Axios as the main technologies for efficient management of blood bank operations that can improve the quality of service and satisfaction of the users. Gupta, et al (2022), show the benefit of implementing data analytics in blood bank management through a Comprehensive Review of Data Analytics Techniques in Online Blood

Bank Management Systems that explores the application techniques such as data mining, machine learning and predictive analytics to optimize blood inventory, predict donation patterns and improve decision-making processes in online blood bank management system—carried out a comprehensive review on a comprehensive review of data analytics techniques in online blood bank management systems.

**Analysis of Existing System**

Hossain & Ferdous (2021), developed an Online Blood Bank Management System Using PHP, HTML, CSS, JavaScript and MySQL to eradicate traditional blood bank management systems. This web application helps a user to find and contact people donating blood in a specified area through phone numbers or email addresses. The user sees the location of other users on the map and gets a notification in case the blood group matches the required blood type. Nearby hospitals can be traced and accessed. This system is limited to only fresh donors and in healthcare management, there is a need for blood to be stored for emergency cases. The existing system did not consider the need for donors to voluntarily register and donate blood for storage keeping for future use rather it only considers the current need for it when a patient is in emergency need of it and still leaves a majority of the donor activity to admin. The design plan adopted the Incremental Model Development Lifecycle methodology to achieve multiple stand-alone modules. This process ensures that each component is developed and tested independently, ultimately leading to a more robust and adaptable system but not flexible. The methodology only suits relatively stable requirements where each phase is clearly defined.

**Disadvantages of the Existing System**

- The system is complex and slower due to the use of incremental model development lifecycle
- Low scalability of the system
- Tedious communication between donors and recipients
- Poor user interface and user-friendly system

**Analysis of the Proposed System**

The proposed system took a significant step forward in the efficient and responsible management of a critical aspect of the Nigeria healthcare system which is blood bank management, using web-based to enhance the flow of blood transfusion between donors, patients and blood bank admin. The system uses populated data containing the address of every registered blood bank into the database to give donors and patients seamless access to nearby blood banks by filtering the user chosen Local government area and then querying the system for blood banks within that chosen address and this will aid quick access and support emergency cases. The system allows blood bank admin to register their blood bank into the system for location management and access to populate their blood bank data and manage donors/patient requests. The GIS/GPS technology was not considered because here in Nigeria the accuracy of the Google map is not yet up to 70% and not everyone has the knowledge to read and under this digital map. The system with further advancement will incorporate other users like the laboratory department and verification system. The system is developed using technologies that allow adaptability and flexibility due to the possibility of new requirements in the future. The technologies used for the development of the system that could support and give a meaningful result to the problem were HTML5, CSS, JavaScript for the frontend (user interface) and PYTHON-DJANGO for the backend and scripting language, MYSQL for the database and Agile methodology for flexibility, scalability, and adaptability base on technology advancement in the healthcare sector.

**Advantages of the Proposed System**

- Improved proficiency in blood stock management aiding blood bank staff to know the blood quantity available and regular updates for blood stock
- Well-organized donor and blood stock management functions of the blood bank, keeping proper records of donor and blood details
- Easy access to different blood types and Improved communication
- Scalability, flexibility and adaptability of the system due to the use of the Agile method and enhanced user-friendliness

**Methodology**

The methodology adopted for the proposed system is **Agile Methodology** using the **Adaptive Software Development (ASD) Method**. The ASD is a flexible approach to the Agile development of complex software systems that encourages emergence, continuous testing and deeply integrated user feedback and collaboration. Adaptive Software Development (ASD) aims to enable effective adaptation of requirement changes and advancement of product needs with lightweight planning and continuous learning. The ASD is a results-oriented framework and places more emphasis on the quality of outcomes than the tasks performed to achieve them. It is mission-driven, Feature-based, Iterative, Time-boxed, Risk driven and Change Tolerant. There majorly three (3) phases in Adaptive Software Development (ASD). These phases reflect the dynamic nature of ASD, explicitly replacing Determinism with Emergence.

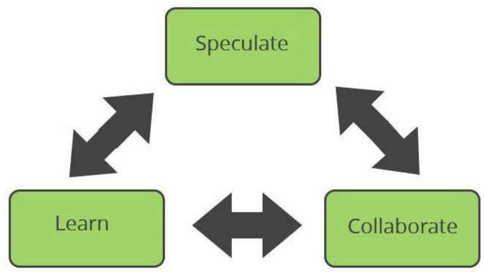



*Figure 1 Agile -Adaptive Software Development Methodology for The Proposed System*

**I. Speculate**

The Speculate phase is a stage in software development where the requirements, plan, and risks are determined. It is called speculate because the outcome is uncertain and may change due to unforeseen factors. The requirements are the needs of the users and stakeholders, the plan is the outline of the features, timeline, and resources, and the risks are the potential challenges or problems that may arise. The speculate phase is repeated in each iteration, which is a cycle of exploration and experimentation

**II. Collaborate**

The collaborate phase is the second phase, where the project team cooperates to produce the software product based on the plan and the requirements. During the collaborate phase, the team will work with stakeholders to refine the requirements and to develop a more detailed plan for each iteration. The team will then start building working software and delivering it to stakeholders for feedback. The collaborate phase is a highly collaborative process. The team should be open to feedback from stakeholders and should be willing to make changes to the plan as needed. The collaborate phase has four activities**:**

- Communication and teamwork
- Feature-driven development
- Quality assurance
- Risk management

**III. Learn**

The learn phase is all about reflecting on the work that has been done and adjusting the plan as needed. At the end of each iteration, the team should reflect on what went well and what could be improved. This information should be used to update the plan for the next iteration. The learn phase is important because it allows the team to continuously improve the development process.

**System design**

**The Propose System Architecture**

The proposed system architecture is a conceptual blueprint that defines how the system will be structured and how it will behave. It is a formal representation of the system, organized in a way that allows users to understand the different components and the interactions between them to achieve the set goals. The architecture is a three tiers layer architecture with the presentation tier which encompasses the user interface elements and the web applications that users interact with, the application tier serves as the intermediary between the presentation tier and the data tier and it handles all the logic, and data tier manages the persistent data storage for the system.

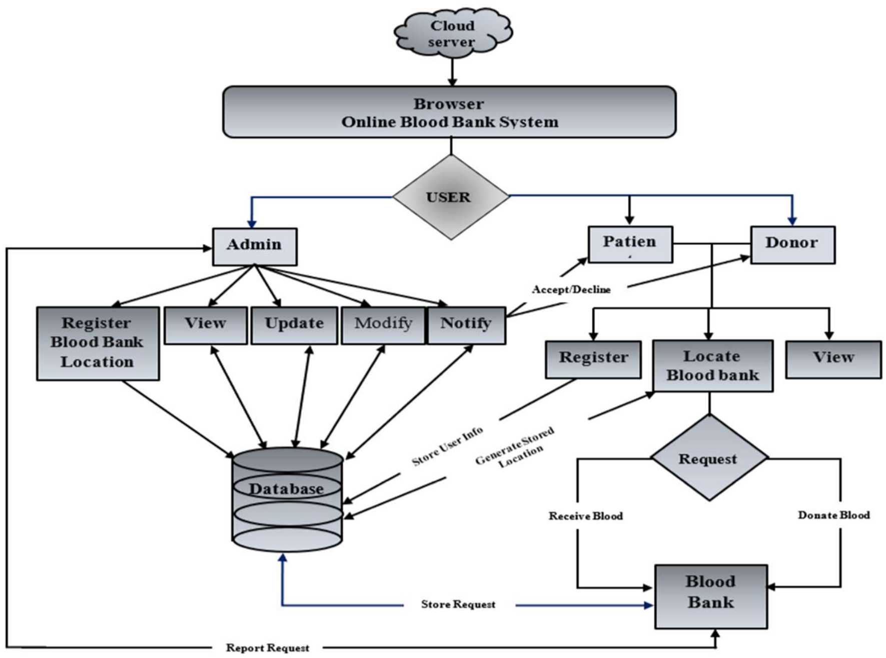



*Figure 2: Architecture of The Proposed System*

**Proposed System Data Flow Diagram**

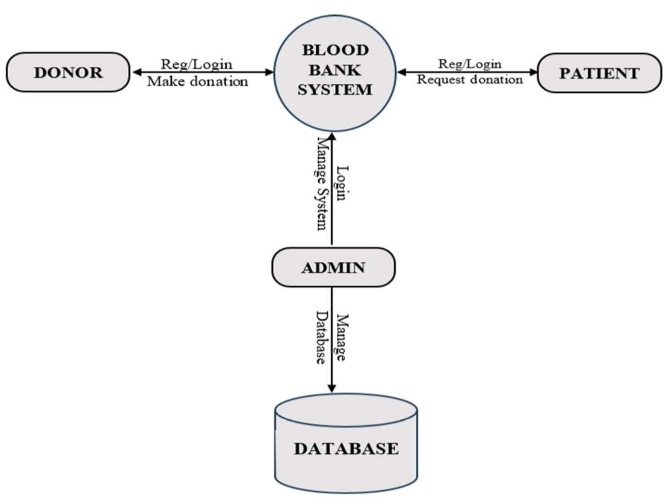



***Figure 3*** *Proposed System DFD level 0*

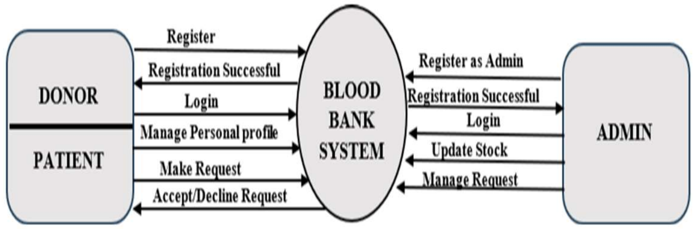



*Figure 4 Proposed System DFD Level 1*

**Use-Case Diagram**

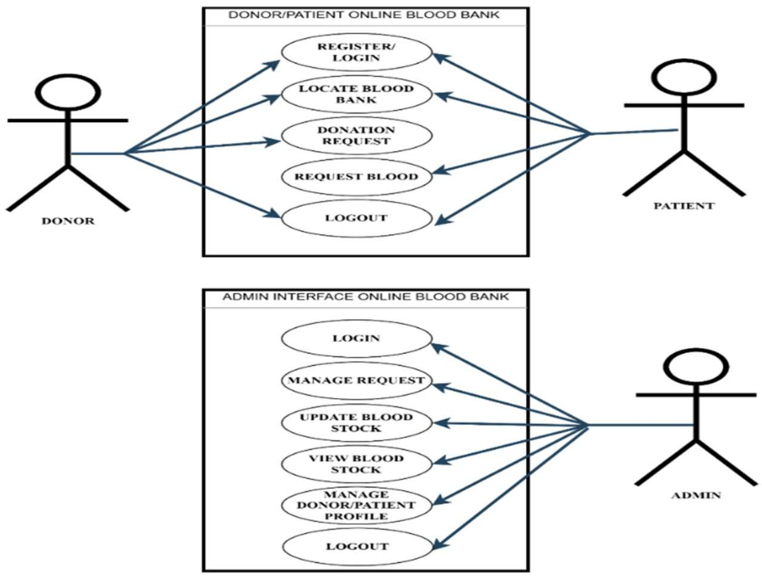



***Figure 5:*** *The Proposed system Use-case diagram*

**The System Flowchart:**

It is a high-level overview of the system that showcases the sequence of the flow of data through the system and how decisions affect this process.

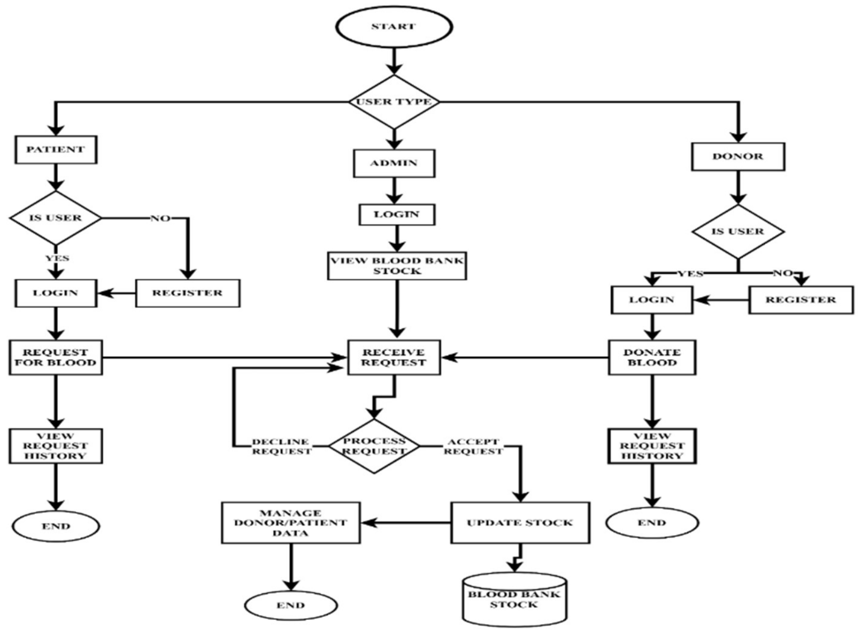



*Figure 6:* *The Proposed system flowchart diagram*

### Entity-Relationship Diagram (ERD) of Proposed System

An Entity-Relationship Diagram (ERD) is a graphical representation of the entities and their relationships in a database. ERDs are used to design and visualize relational databases, which are the most common type of database used today. MYSQL which is chosen for the system is also a relational database. To represent the database, we will be using Crow's foot notation to graphically display the entities and their relationships.

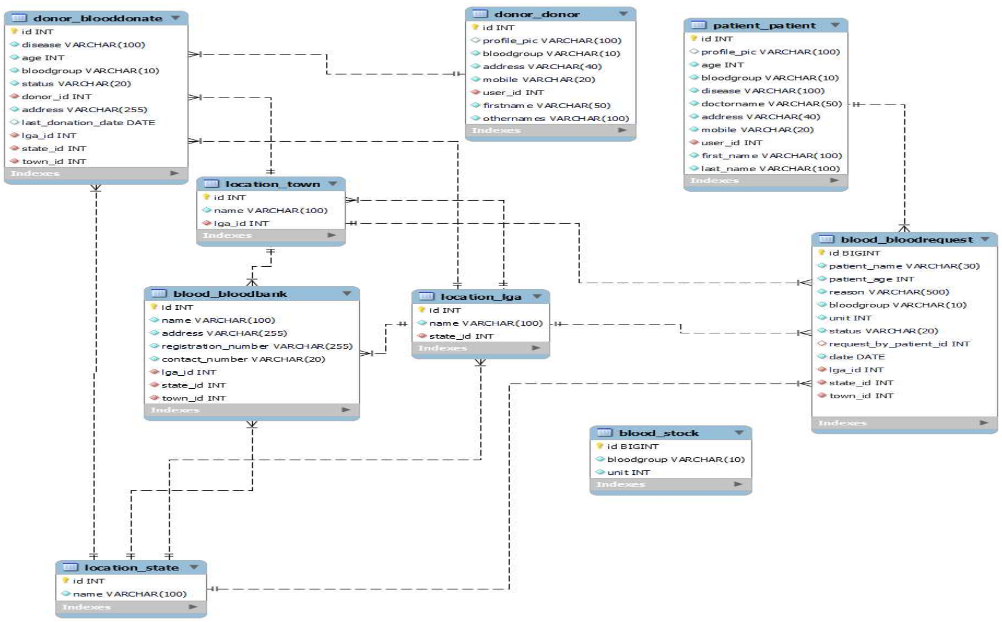



*Figure 7: The Proposed system Entity Relationship Diagram (ERD)*

### System Requirement

1. **Server requirement**
   i. Operating system: Windows (7, 8, 10, 11) Mac OS, Linux OS, Android OS, iOS etc.
   ii. **Server Resources:** Multi-core CPU, and RAM of 2 GB and above.
   iii. **Database Management System:** A relational database management system (RDBMS) like MySQL Workbench or Oracle.
   iv. **Web Server**: Apache or Microsoft Internet Information Service (IIS).
2. **Hosting Environment**
   i. **Cloud Hosting Platform**: Amazon Web Services (AWS), Google Cloud Platform (GCP) or Microsoft Azure.
3. **Web Application Requirements**
   i. **Programming Language:** Python, CSS, HTML or JavaScript (jQuery/AJAX)
   ii. **Framework**: Python Django
   iii. **Environment:** Visual Studio code, Notepad++, PyCharm, or anaconda.
4. **Client-Side Requirements:**

i. **Interface:** Desktop computer, Laptop, Tablet, Smartphone

ii. **Browser:** Chrome, Windows Explorer, Firefox, Opera browser, UC browser etc.

5. **Security Requirements:** SSL Certificate, User Authentication and Authorization
6. Bandwidth and internet connection

## Results

**Homepage:** The homepage is the first page (welcome page) that every user of the system will interact with before login/registration. It has the patient link, donor link and admin link to direct users to their respective page.

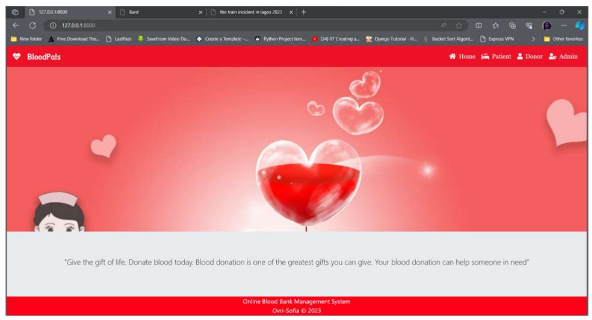



*Figure 8: Homepage*

**Admin Registration Page:** Since the admin is a super user, they cannot register into the system like donor and patient but can only register into the system from the back-end which can only be done by the system engineer/developer.

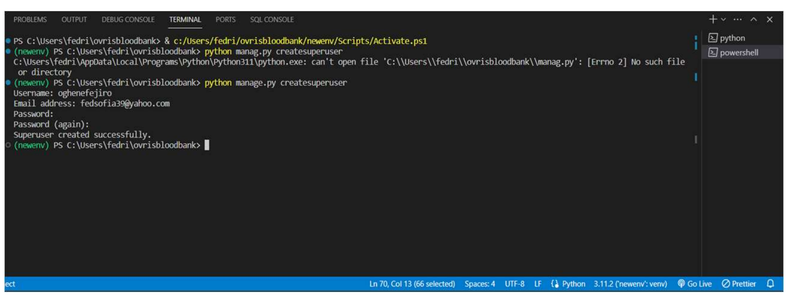
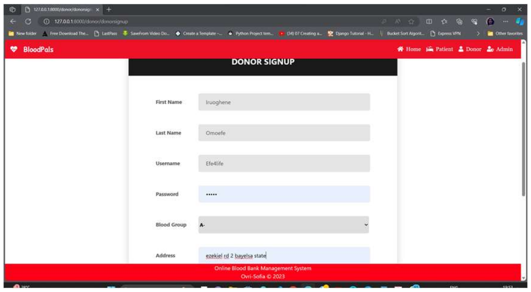



*Figure 9: Admin Registration*

**Registration pages:** It allows both donors and patients to register into the database individually, and easily gain access to make requests.



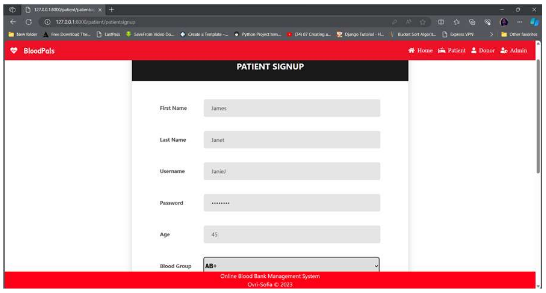
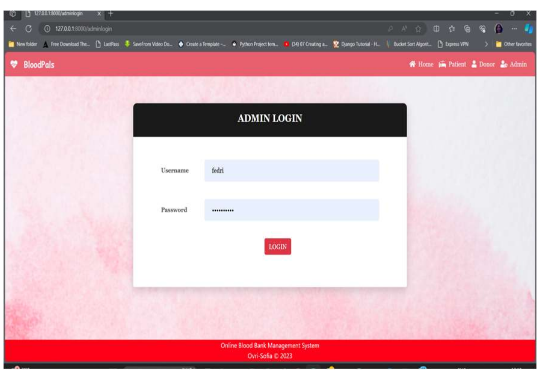




***Figure 10: Donor Registration Page*** ***Figure 11: Patient Registration Page***

**Login pages:** The below interfaces will authenticate users and grant them access to their respective pages to perform their respective tasks or allow donor and patient users to navigate to the registration page if they are not a register user yet.



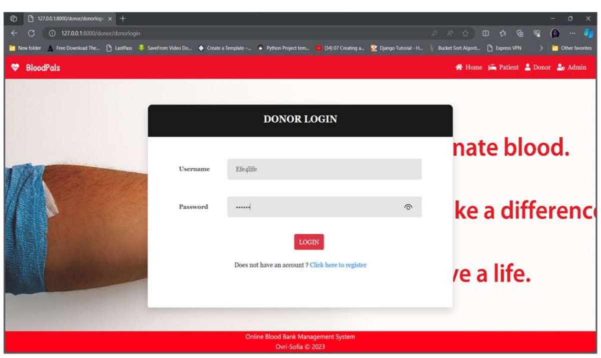




*Figure 12: Admin login page* *Figure 13: Donor login page*

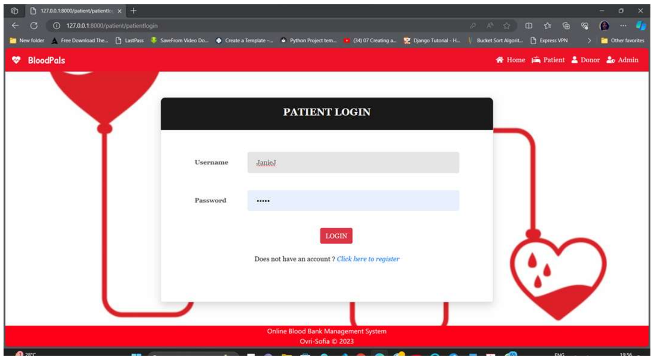



***Figure 14:*** *Patient login page*

**Admin Dashboard**: The dashboard displays the bloodstock available in the blood bank that can only be inputted by an authorized admin and the total number of donors and requests the system has and has attended to. It has a sidebar for other activities.

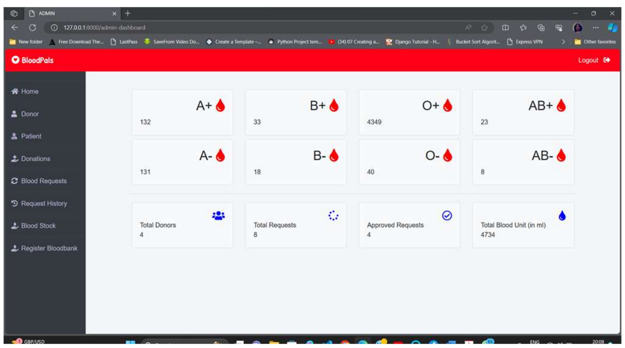



*Figure 15: Admin dashboard*

**Admin Blood Request:** On this page, the admin views all requests for blood recipients and acts on them either by accepting or declining based on the level of blood stock available.

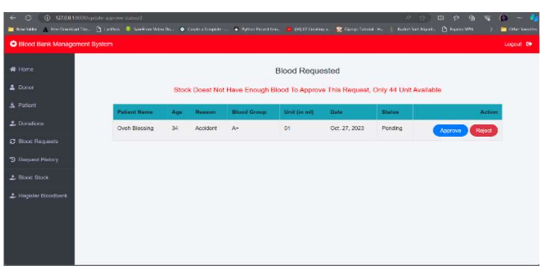



***Figure 16****: Blood request management page*

**Donor and Patient Management Page:** The Admin can edit the donor and patient details if in the case the donor register with a wrong blood group or have a change of address. The admin can also delete donors if need be.

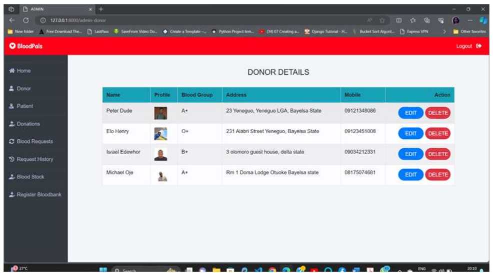



***Figure 17:*** *Donor Management Page*

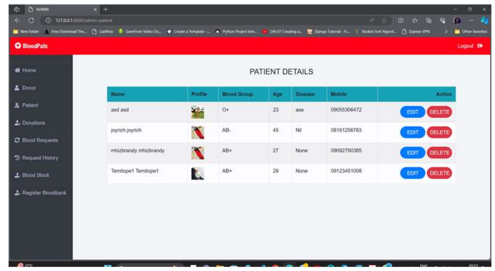



***Figure 18:*** *Patient Management Page*

**Donation Request Management Page:** The admin request management page, receive request for blood and can choose to accept or decline using any the buttons on the action column depending on the level of blood stock available and also if the blood available is not up to requirement and the admin fails to check, the system can automatically decline the request when the blood available is up to what is needed.

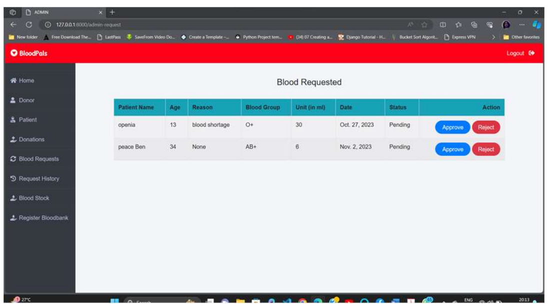



***Figure 19****: Donation Request Management Page*

**Request History page:** This page displays the history requests from patients which was neither approved nor rejected base on certain factor the admin consider

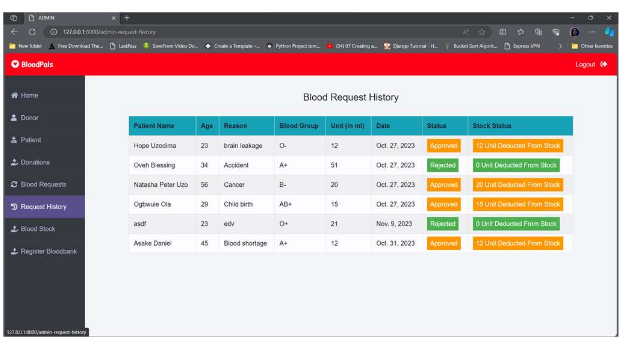



***Figure 20:*** *Request History page*

**Stock Dashboard:** The Admin Blood Stock Dashboard display the available units of blood for each blood group and also enables the admin to add (update) new blood stock available in the blood bank after a donation is made successfully.

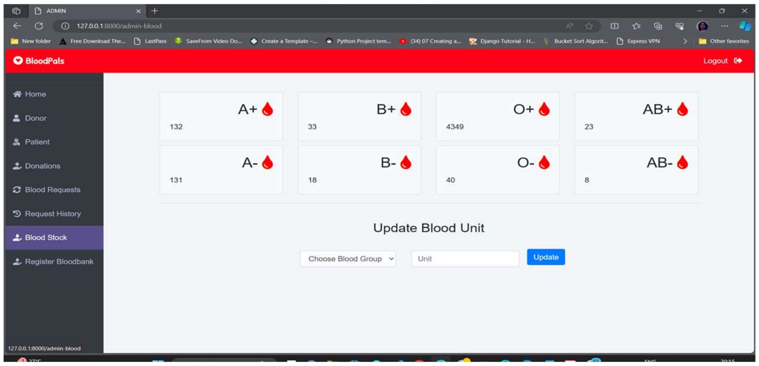



***Figure 21:*** *Blood Stock Dashboard*

**Donor Dashboard:** The donor dashboard is the first page a donor user interacts with after login. It displays the summary activity of the donor and there is a sidebar menu that allows the donor to navigate and perform other tasks like sending an application for blood donation.

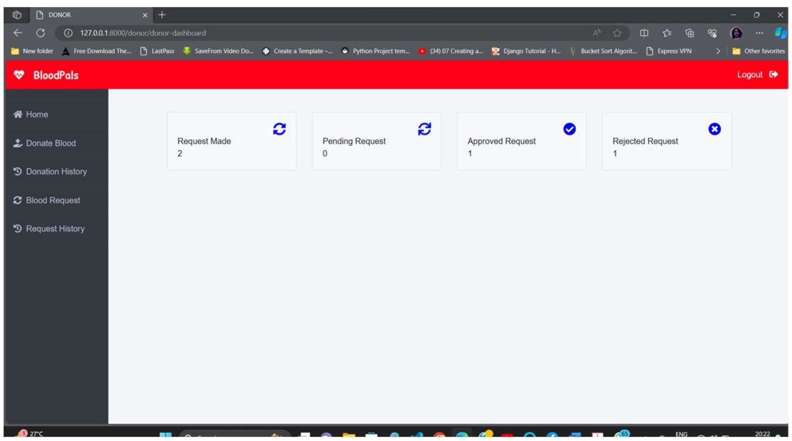



***Figure 22:*** *Donor Dashboard*

**Donor Request form:** The donor request form page performs the streamlined search of nearby blood banks by allowing a donor to select their current state, LGA, and town and then the system will query the database to generate all blood banks within the selected LGA and Town and allow a donor to make a pick on which blood bank they can send their request

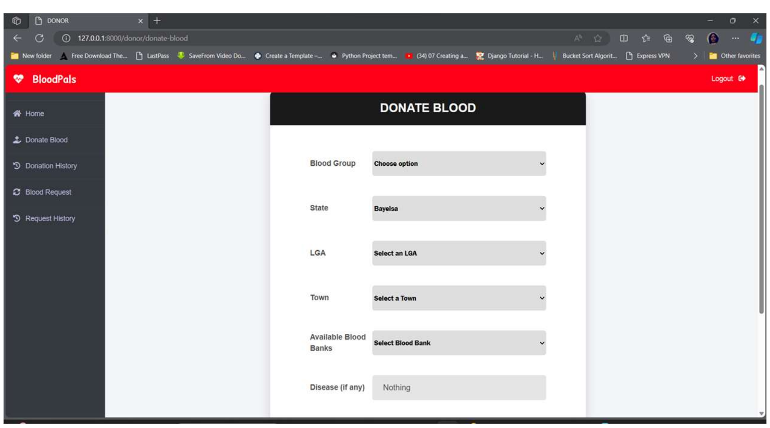


*Figure 23: Donation Request form page*

**Donor's History Chart:** The donor history dashboard displays a list of requests the donor has made so far including the requests that were declined or accepted. Here donor can check if their request was accepted or rejected to know if they can proceed to the blood bank to donate.

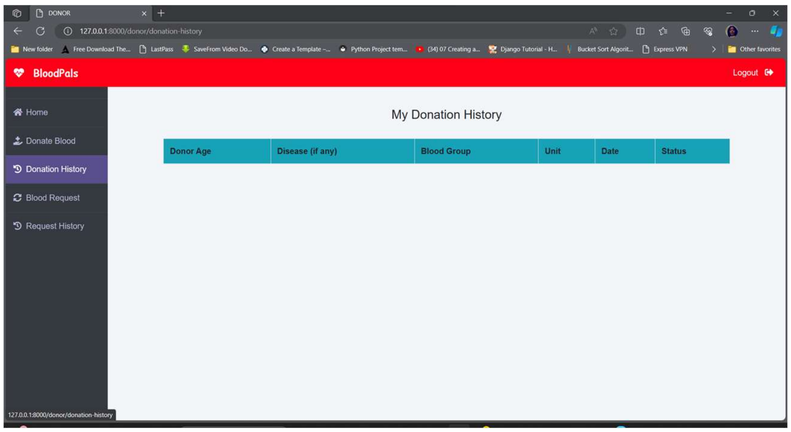



*Figure 24: Donor's History Chart*

**Patient Request Form:** The patient request form is the form the patient fills and sends to the admin to request units of blood with reasons for the request. The patient selects an LGA and the town field filters the town database and generates towns in the LGA. The blood bank field uses the LGA and Town selected by the patient to query the database and generate all blood banks in that given LGA and Town for the patient to select the blood bank closer to the required blood type and send their request for admin to approval and then the patient proceeds to the physical collection of the blood product request else if the request is declining the patient can send another request to the next blood bank.

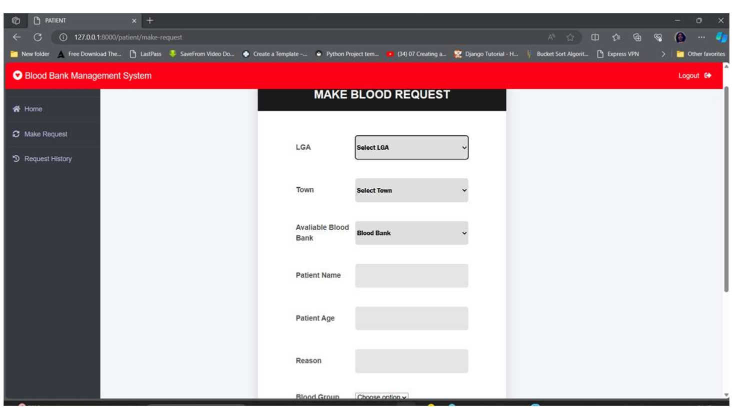



*Figure 25: Patient's Blood Request form*

## Conclusion

The design and implementation of an online blood bank management system will usher blood bank management in Nigeria into a digital era for effectiveness and efficiency. The study harnessed a range of cutting-edge technologies which gives room for further evolvement to meet the trending technology. The world today operates in digital mode, manual storage of data is all fading out and manually locating places also. With the advancement of technology, one can sit at home and get the address of any possible location they wish to go. Saving a life requires proper timing and this study has the potential to help with timing to get blood for donation and also timing to help donors locate blood banks. The successful implementation and evaluation of the system underscore its potential to significantly impact and improve the blood bank management landscape for further advancement in the field using the study of computer science and informatics to advance healthcare institutions.

**Contribution to Knowledge**

The contribution to knowledge from the development of the online blood bank management system is significant. The system has proven to be effective and efficient in addressing real-time implementation of the blood bank management system. It is flexible, user-friendly, and easily adaptable to the changing world of inventory management. It also simplifies the process of locating nearby blood banks with database query functionalities.

**Recommendation**

The researchers recommend a collaboration with other healthcare sectors and their stakeholder to establish industry standards for blood bank management systems that can promote interoperability and best practices in the field, for example, the laboratory sector of healthcare institutions for sample testing and donation recommendation.